Conversion of Borneol to Camphor

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Introduction

In this experiment borneol was converted to camphor. In this process, borneol, an alcohol, is oxidized into camphor, a ketone. This is the last step in the process of the conversion of camphene to camphor. This conversion was achieved though simple distillation. Bleach is added as an oxidate. This produces a crude camphor which is purified by sublimation. Acidic residuals are eliminated by adding sodium bicarbonate to neutralize the solution. The organic solution is dried with sodium sulfate to remove any microscopic water before evaporating the DCM over a water bath.

Materials and Methods

The experiment begins by adding 0.3 g of the borneol product with 0.3 mL of glacial acetic acid to a long-neck 10 mL round-bottom flask. The racemic mixture is dissolved in 1 mL of acetone and a boiling chip is added. A reflux with the distillation column is assembled and the apparatus is placed in a water bath with a temperature between 40-50°C. If the borneol does not dissolve, 0.5 mL of acetone is added. After all reagents dissolve, 6 mL of bleach is added. This addition is done through the top of the condenser and is dropwise over 35 minutes. After each addition the mixture is given a gentle swirl. After the last addition, the reaction is allowed to continue for 15 minutes. After the mixture cools to room temperature, the camphor is extracted with two 2 mL portions of DCM using a centrifuge tube. The mixture is then neutralized by 2 mL of saturated sodium bicarbonate. The organic layer is washed sequentially with 2 mL of 5% sodium bisulfite and 2 ml of water. The organic solution is dried with anhydrous sodium sulfate. The DCM is evaporated over a water bath. The collected crystals are weighed, and percent yield is calculated. The melting point is compared with the range published in literature.

Results

Weight of Watch Glass	31.678 g
Weight of Watch Glass + Actual Yield	31.710 g
Actual Yield (Weight of Product)	0.032 g
Theoretical Yield	0.296 g
Percent Yield	10.81% yield

Table 1. Actual Yield, Theoretical Yield, and Percent Yield.

	Reactant (Borneol)	Product (camphor)
Amount Used	0.3 g	--
Yield	--	0.032 g
Molar Mass	154.25 g/mol	152.23 g/mol

Table 2. Data used for calculations

$$\frac{0.30\text{ g reactant}}{} \text{ x } \frac{1\text{ mol reactant}}{154.25\text{ g reactant}} \text{ x } \frac{1\text{ mol reactant}}{1\text{ mol product}} \text{ x } \frac{152.23\text{ g product}}{1\text{ mol product}} = 0.296\text{ g product}$$

$$\frac{0.032\text{ g}}{0.296\ g} \text{ x } 100 = 10.81\%\text{ yield}$$

Discussion

In this experiment, borneol was converted to camphor through oxidation. Assessment of the reaction determined the product to be camphor. This assessment included examining the reaction formula as well as certain steps in the procedure. Adding, sodium bicarbonate ensured no glacial acetic acid in the product and anhydrous sodium sulfate removed any water from the product. The final step to remove the DCM ensured that the only material left was the product, camphor. The percent yield determined was 10.81. This leads to the belief that there were multiple sources of error or loss of product. One of these factors could be not allowing the reaction to complete. If the mixture was not allowed to sit for 15 minutes after adding the bleach it is possible that not all

the borneol was converted to camphor, which would decrease percent yield. Another way percent yield could be affected is by not allowing layers to completely settle before extracting. This would allow some of the organic product to get lost in the aqueous layer and decrease percent yield.

Lab Performance

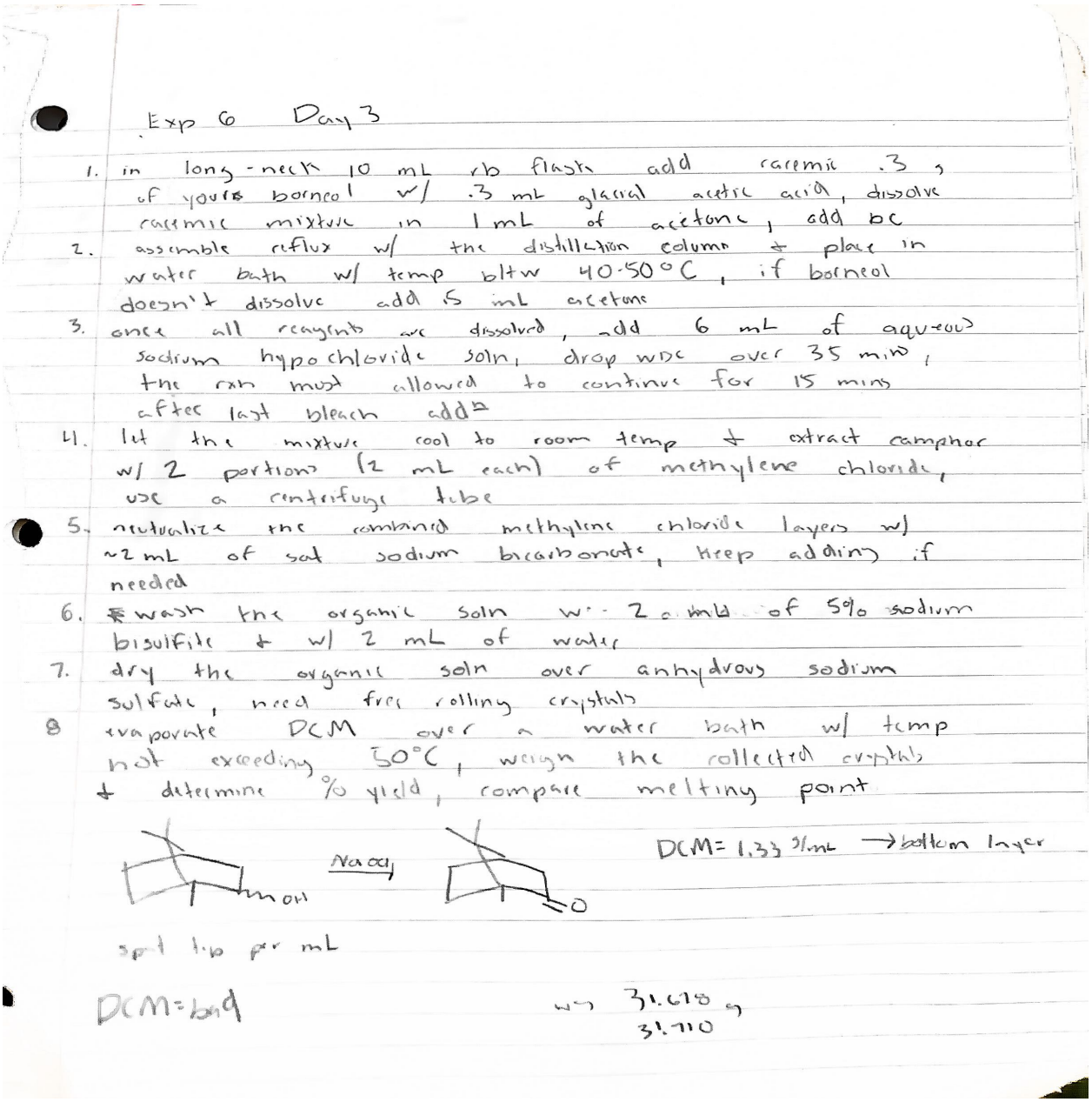
Exp 6 D3

Name: *Section:* *Room:* *Hood:*

Wearing personal protection equipment—this evaluation records your consistency with wearing the appropriate clothes for lab (long pants, closed toe shoes, and tops with sleeves) as well as the required laboratory safety equipment of apron, goggles, and gloves.	○ ○ ○ ○ ○ (5pts)
Safe working habits—as the GTAs do the lab rotations, they will record your work ethic and whether you follow our safety rules and regulations concerning tied hair, tied apron, safety goggles on at all times, and no food and drinks.	○ ○ ○ ○ ○ (5pts)
Following our laboratory safety technique work procedures—evaluation of operation of our laboratory equipment—water sinks in hood (1pt), vacuum and air lines (1pt), sash position (1pt), and heating elements (1pt). It is important that after every experiment you shut down the hood and make it ready for the section after yours (5pts).	○(1pt) ○(1pt) ○(1pt) ○ (1pt) ○ (5pts)
Glassware operation—safe handling of our glassware equipment. This means cleaning (1pt), maintaining the drawer inventory (5pts), and proper experimental setup (5pts).	○(1pt) ○ (5pts) ○ (5pts)
Working independently or equal partner participation during the entire lab.	○ ○ ○ ○ ○ (5pts)
Good work behavior—you work in a dynamic environment; therefore, creating distractions through noise and disobedience will not be tolerated. No cell phone use, listening to loud music, or working/distracting at a non-assigned workspace.	○ ○ ○ ○ ○ (5pts)
Maintain clean and organized personal space.	○ ○ ○ ○ ○ (5pts)

GTA notes:

GTA signature: _____ *Final Performance Grade:* _____



References

Keller L, Joseph K, and Miguel S. 2022 Fall. CHEM 2071 Organic Chemistry II Lab Manual [lab manual]. Auburn (Al): Auburn University.